

Your Thesis Title Here

Your Name Here

Master's Thesis

2026

Supervisor: Supervisor 1
Supervisor 2
Supervisor 3

Examiner: Examiner



University of Stuttgart
Institute of Flight Mechanics and Controls

Task Description for the Master's Thesis

of Student Name

Working Title

Motivation:

A few sentences about the motivation for this thesis. This should explain why the topic is interesting and relevant.

Objective:

A brief description of the main objective of the thesis.

Tasks:

- Task 1
- Task 2
- Task 3

Start:

31.12.2048

Supervisor:

Supervisor 1, Supervisor 2, Supervisor 3

Signature of Supervisor

Signature of Student Name

Erklärungen für Abschlussarbeiten am iFR

Erklärungen bei Ausgabe des Themas der Arbeit

Mir ist bekannt, dass die Arbeit im Online-Katalog der Bibliothek und auf der Institutswebseite erfasst wird, was eine dauerhafte, weltweite Sichtbarkeit der bibliographischen Daten der Arbeit zur Folge hat.

- ☐ Ich stimme der Nennung meines Namens und einer Kurzfassung der Arbeit zu diesem Zweck zu.
- ☐ Ich räume der Universität Stuttgart, Institut für Flugmechanik und Flugregelung, an dieser Arbeit und den im Rahmen dieser Arbeit entstandenen Ergebnissen ein kostenloses, zeitlich und räumlich unbeschränktes, einfaches Nutzungsrecht ein. Dieses erstreckt sich auf sämtliche bekannte Nutzungsarten und umfasst neben dem Recht auf Nutzung in Forschung, Lehre und Studium insbesondere das Recht der Vervielfältigung, Verbreitung und Änderung sowie das Recht der öffentlichen Zugänglichmachung.

Ort, Datum

Your Name Here

Erklärungen bei Abgabe der Arbeit

Hiermit versichere ich, dass ich diese Arbeit selbständig mit Unterstützung der Betreuer angefertigt und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe. Fremde Beiträge habe ich als solche gekennzeichnet. Die Arbeit oder wesentliche Bestandteile davon sind weder an dieser noch an einer anderen Bildungseinrichtung bereits Gegenstand einer Prüfungsleistung gewesen und wurden nicht ohne Zustimmung der Prüferin bzw. des Prüfers bereits veröffentlicht. Ich erkläre weiterhin, bei der Erstellung der Arbeit die einschlägigen Bestimmungen entsprechend den Regeln guter wissenschaftlicher Praxis¹ eingehalten zu haben. Ich versichere, dass alle gedruckten Exemplare mit der digitalen Abgabe der Arbeit übereinstimmen.

Sollte ich der Universität Stuttgart, Institut für Flugmechanik und Flugregelung, Nutzungsrechte an dieser Arbeit und den im Rahmen dieser Arbeit entstandenen Ergebnissen eingeräumt haben, versichere ich, eventuell erforderlich gewordene Zustimmungen der Urheber zur Nutzung fremder Beiträge in meiner Arbeit eingeholt zu haben. Ich erkläre mich für den Fall der Verletzung der Rechte Dritter durch meine Arbeit bereit, der Universität Stuttgart einen daraus entstehenden Schaden zu ersetzen bzw. die Universität Stuttgart auf deren Aufforderung von eventuellen Ansprüchen Dritter freizustellen. Zusätzlich zu den vom Prüfer geforderten Exemplaren, übertrage ich der Universität Stuttgart das Eigentum an einem von mir der Bibliothek des Instituts für Flugmechanik und Flugregelung kostenlos zur Verfügung gestellten Exemplar dieser Arbeit.

Ort, Datum

Your Name Here

¹Siehe "Leitlinien zur Sicherung guter wissenschaftlicher Praxis" der DFG bzw. "Satzung der Universität Stuttgart zur Sicherung der Integrität wissenschaftlicher Praxis und zum Umgang mit Fehlverhalten in der Wissenschaft".

Abstract

In here, you should briefly describe the thesis content in the primary language.

Kurzfassung

In here, you should briefly describe the thesis content in the secondary language.

Acknowledgements

Contents

Task Description	iii
Erklärungen	v
Kurzfassung / Abstract	vii
Acknowledgements	ix
List of Figures	xiii
List of Tables	xv
1. Introduction	1
1.1. About the template	1
1.1.1. Key Features	1
1.1.2. Paragraph Indentation	1
1.1.3. Paragraph Breaks	1
2. Literature Review	3
2.1. How to use the nomenclature	3
2.2. How to cite from the bibliography file	4
2.3. Important Notes	5
3. Developed work	7
3.1. General Guidelines	7
3.2. Mathematical Equations	7
3.3. Tables and Data Presentation	7
3.4. Figures and Visual Elements	8
3.4.1. Multiple Images with Shared Caption	8
4. How to add formatted code	9
4.1. Basic Example	9
4.2. Creating Custom Formats	9
A. First Appendix	11
A.1. Heading in the Appendix	11
B. Second Appendix	13

List of Figures

3.1. Block diagram of a control system	8
3.2. Controller with pre-filter structure	8
3.3. Short description	8

List of Tables

2.1. Common biblatex citation commands	5
3.1. Controller Design Parameters	7

1. Introduction

This chapter serves as a guide to using this thesis template effectively. It covers the template's features, basic typographic rules, and best practices for structuring your document. Whether you're a new user or an experienced $\text{\LaTeX}$ writer, these guidelines will help you produce a professional and consistent thesis.

1.1. About the template

The iFR thesis template is designed to streamline the writing process while ensuring professional formatting and adherence to academic standards. Beyond its clean and polished appearance, this template includes comprehensive examples and built-in functionality for common thesis elements.

1.1.1. Key Features

This template comes with ready-to-use examples for:

- How to use the nomenclature
- How to include images in different layouts
- How to cite from the bibliography (.bib) file
- How insert a formatted piece of code

For comprehensive information beyond this introduction, please consult the `README.md` file included with this template. This document contains essential information about:

- How to build your final pdf document
- How to set up VSCode

1.1.2. Paragraph Indentation

First paragraphs after headings (chapters, sections, subsections) **should not be indented**. This is a standard typographic convention that helps visually separate section breaks from normal paragraph breaks. Subsequent paragraphs within the same section should be indented.

1.1.3. Paragraph Breaks

Never force line breaks manually using `\\` or `\newline` between paragraphs. Instead, separate paragraphs with a blank line in your source code. This approach:

- Allows $\text{\LaTeX}$ to optimize line and page breaks
- Ensures consistent paragraph spacing throughout the document
- Makes your source code more readable and maintainable
- Prevents formatting errors when the document is recompiled

2. Literature Review

2.1. How to use the nomenclature

The nomenclature entries can be added from anywhere in the document and must be assigned to one of the four categories:

A Abbreviations,

B Latin Letters,

C Greek Letters,

D Indices

Entry Format

Each nomenclature entry follows this syntax:

```
\nomenclature[Category]{Symbol}{Description}
```

Where Category is one of: A (Abbreviations), B (Latin letters), C (Greek letters), or D (Indices).

Examples

1. Abbreviations (Category A):

```
\nomenclature[A]{KOS}{Koordinatensystem}
```

This creates an entry for "KOS" with the description "Koordinatensystem".

2. Latin letters (Category B):

```
\nomenclature[B]{\boldsymbol{r}}{\Vektor vom Lasthaken zum Schwerpunkt}
```

3. Greek letters (Category C):

```
\nomenclature[C]{\Omega}{Drehgeschwindigkeit (\si{\radian\per\second})}
```

This adds the Greek letter Omega with its description. Some authors like to add the quantity's unit in the nomenclature as well. Here, this is done in parentheses as part of the description.

4. Indices (Category D):

```
\nomenclature[D]{$_g(\cdot)}{\text{Geodätisches Koordinatensystem}}
```

This defines the subscript 'g' used in equations throughout the document.

2.2. How to cite from the bibliography file

Your references are stored in a bibliography file (with extension .bib). In this document template, the bibliography file is located at `src/bib/references.bib`. Open this file to see how bibliography entries are structured.

A typical bibliography entry looks like this:

```
@BOOK{Johnson1994,  
  title = {Helicopter theory},  
  publisher = {Dover Publications},  
  year = {1994},  
  author = {Johnson, Wayne},  
  address = {Mineola, NY}  
}
```

The entry starts with the document type (`@article`, `@book`, `@inproceedings`, etc.) followed by a citation key (Johnson1994) that you'll use to cite this source in your document.

There are two main approaches to building your bibliography:

Manual Entry

For small bibliographies, you can create entries by hand. This gives you complete control but requires attention to detail. Follow the formats shown in your `references.bib` file.

Automatic Entry (Recommended)

For larger bibliographies, use reference management tools to automatically add entries:

- **Reference managers:** Tools like Zotero, Mendeley, or EndNote can export collections as BibTeX files.
- **DOI to BibTeX:** Use websites like <https://doi2bib.org> to convert DOIs to BibTeX entries.
- **CrossRef:** Visit <https://www.crossref.org> to look up and export citations.

Always check automatically generated entries for consistency and completeness. Pay special attention to:

- Proper capitalization (protect with braces, e.g., NASA)
- Special characters and accents (use proper $\LaTeX$ encoding)
- Complete page numbers and DOIs when available

Once you have entries in your bibliography file, you can cite them using various commands. This template uses `bibtex` with the `numeric` style, which provides flexible citation options:

Table 2.1.: Common biblatex citation commands

Command	Example Usage	Result
<code>\cite{key}</code>	<code>\cite{Fichter2009}</code>	[Fichter2009]
<code>\textcite{key}</code>	<code>\textcite{Fichter2009}</code>	Textual: Fichter2009
<code>\citeauthor{key}</code>	<code>\citeauthor{Fichter2009}</code>	Only authors: Fichter2009
<code>\citeyear{key}</code>	<code>\citeyear{Fichter2009}</code>	Only year: Fichter2009
<code>\cite[42]{key}</code>	<code>\cite[42]{Fichter2009}</code>	With page number: [Fichter2009]
<code>\cite[see][]{key}</code>	<code>\cite[see][42]{Fichter2009}</code>	With pre/post notes: [Fichter2009]
<code>\cite{key1,key2}</code>	<code>\cite{Fichter2009,Johnson1994}</code>	Multiple: [Fichter2009, Johnson1994]

2.3. Important Notes

- Nomenclature entries will automatically appear in the appropriate subsection (Abbreviations, Latin letters, etc.) based on their category.
- For this document, the bibliography items are sorted in the order they are cited in the text.
- A single run of `pdflatex` is generally neither sufficient to correctly update nomenclature and bibliography lists nor add new citations in the text. Use the `latexmk` tool (as with the `full-build` recipe provided in this repository) whenever you add or modify entries to ensure they are properly generated and included in the document.

3. Developed work

3.1. General Guidelines

- **File formats:** Use PDF or EPS for vector graphics (diagrams, plots), PNG or JPEG for photographs.
- **Cross-referencing:** Always use `\label` and `\ref` for all equations, tables, and figures.
- **Captions:** Provide both short (for list of figures/tables) and long descriptions where appropriate.
- **Placement:** Use `[htbp]` for flexible figure/table placement.
- **Units:** Use proper math mode formatting for units (e.g., m/s^2).
- **Consistency:** Maintain consistent notation throughout the document.
- **Spacing:** Use `\quad` or `\qquad` for spacing in equations.

3.2. Mathematical Equations

When presenting mathematical content, ensure equations are properly formatted and referenced. Use the `equation` environment for single equations and `align` for multiple lines.

$$\dot{\varphi} = \frac{1}{\cos \theta} \begin{bmatrix} \cos \theta & \sin \phi \sin \theta & \cos \phi \sin \theta \\ 0 & \cos \phi \cos \theta & -\sin \phi \cos \theta \\ 0 & \sin \phi & \cos \phi \end{bmatrix} \omega \quad (3.1)$$

Equation 3.1 shows the relationship between angular velocities and Euler angle rates. Always introduce equations in the text before presenting them.

3.3. Tables and Data Presentation

Tables should be clear, self-contained, and properly formatted. Use the Tables Generator tool for assistance. Maintain consistent formatting with your thesis style.

Table 3.1.: Controller Design Parameters

Parameter		Value	Unit
Maximum acceleration	$a_{\max}$	1.0	m/s^2
Maximum angular rate	$q_{\max}$	1.0	rad/s
Maximum bank angle	$\eta_{\max}$	6.0	$^\circ$

Table 3.1 summarizes the key design parameters. Always reference tables in the text and use consistent terminology.

3.4. Figures and Visual Elements

Figures enhance understanding of complex concepts. Include clear captions and proper references. For TikZ diagrams, consider using direct input for better integration. Figure 3.1 illustrates the structure of a typical feedback control system. For vector graphics, PDF format is preferred.

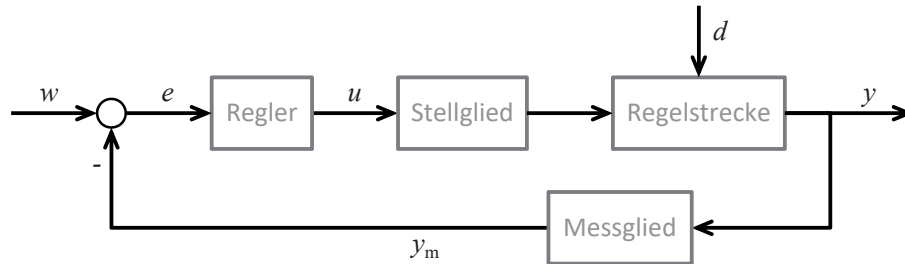


Figure 3.1.: Block diagram of a feedback control system showing the basic components: plant, controller, and feedback loop.

For complex diagrams created with TikZ/PGF, you can include them directly:

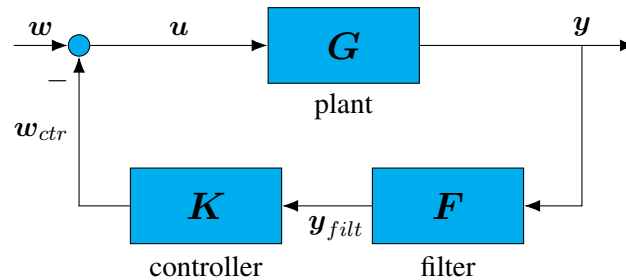
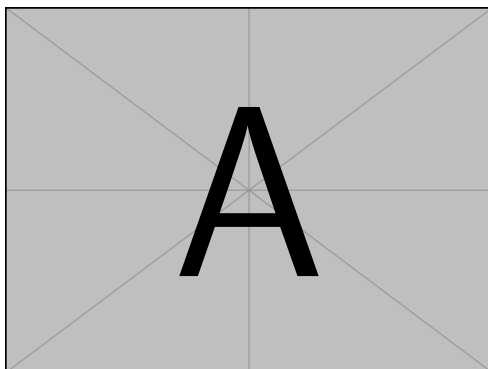


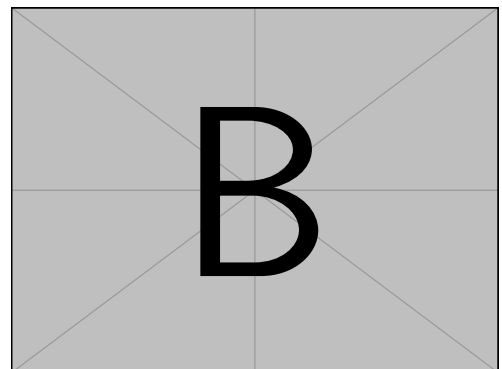
Figure 3.2.: Controller with pre-filter structure

3.4.1. Multiple Images with Shared Caption

When you need to show related images that should share a single caption with individual sub-captions, use the subfigure environment. This is ideal for comparing different views, stages, or aspects of your work.



(a) Front view of the satellite.



(b) Rear view showing connectors.

Figure 3.3.: Long description

4. How to add formatted code

4.1. Basic Example

This is some random MATLAB code. You can reference it just like "Listing ??", or "Listing 4.1".

Example 1: Loading from External File

The following code is loaded from an external MATLAB file, showing only lines 7-20:

Example 2: Inline Code Block

You can also write code directly in your document:

Listing 4.1: Simple MATLAB function example

```
1 function y = myFunction(x)
2     % MYFUNCTION Computes square of input
3     % Input:  x — numeric value
4     % Output: y — x squared
5
6     y = x^2;
7
8     if y > 100
9         disp('Large value detected!');
10    end
11 end
```

4.2. Creating Custom Formats

You can create your own custom formats by defining new styles. Comprehensive documentation and examples for the listings package can be found at: <https://ctan.org/pkg/listings>.

A. First Appendix

Write your appendix content here.

A.1. Heading in the Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

B. Second Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. test